19 **A**, **B** and **C** are three spheres.

The volume of sphere **A** is $125\,cm^3$
The volume of sphere **B** is $27\,cm^3$

The ratio of the radius of sphere **B** to the radius of sphere **C** is 1 : 2

Work out the ratio of the surface area of sphere **A** to the surface area of sphere **C**.

Sphere volumes A:B = 125:27.

Radii A:B = cuberoot(125):cuberoot(27) = 5:3.

B:C radii = 1:2, so radii A:C = 5:6.

Surface areas A:C = 5^2:6^2 = 25:36.

1 mark
radii A:B = 5:3

1 mark
radii A:C = 5:6

1 mark
area ratio 25:36

25 : 36

(Total for Question 19 is 3 marks)

DO NOT WRITE IN THIS AREA
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